Task 1.1: Einstein Summation Convention

Expand the following expressions into their full forms

a) $v_k \boldsymbol{e}_k$,

b) $A_{ij} B_{ji}$,

c) $b_i = A_{ij} v_j$.

a)

$v_k \underline{e}_k$ – Component notation:

Symbolic notation: $\underline{v}$

Vector notation: $\begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$

Full form: $v_1 \underline{e}_1 + v_2 \underline{e}_2 + v_3 \underline{e}_3$

b)

$A_{ij} B_{ji}$ – Coefficient notation

Full form:

$$A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} + A_{21}B_{12} + A_{22}B_{22} + A_{23}B_{32} + A_{31}B_{13} + A_{32}B_{23} + A_{33}B_{33}$$

c)

$b_i = A_{ij} v_j$ – Coefficient notation

Full form:

$$b_1 = A_{11}v_1 + A_{12}v_2 + A_{13}v_3$$
$$b_2 = A_{21}v_1 + A_{22}v_2 + A_{23}v_3$$
$$b_3 = A_{31}v_1 + A_{32}v_2 + A_{33}v_3$$

Fill in the missing indices of the tensor $\boldsymbol{A}$

a) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mm},$

b) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mr},$

c) $A \quad B_{kr} C_{jks} \delta_{ji} + E_{ml} \delta_{ur} F_{lu} = G_{mr}$

a) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mm}$

Free indices:	LHS	RHS
	3	0

- Make all the free indices repeated indices!

$$A_{irs} \delta_{ij} B_{kr} C_{jks} = D_{mm}$$

b) $A \quad \delta_{ij} B_{kr} C_{jks} = D_{mr}$

Free indices:	LHS	RHS
	3	2
	i, r, s	m, r

- Make the free indices that are not in the RHS repeated indices and add any missing free indices!

$$A_{mis} \delta_{ij} B_{kr} C_{jks} = D_{mr}$$

c) $A \quad B_{kr} C_{jks} \delta_{ji} + E_{ml} \delta_{ur} F_{lu} = G_{mr}$

$$A_{msi} B_{kr} C_{jks} \delta_{ji} + E_{ml} \delta_{ur} F_{lu} = G_{mr}$$

Free indices:

LHS 1	LHS 2	RHS
3	2	2
r, s, i	m, r	m, r

Note: With the given information, only the number and letters of the indices of A can be determined, the position of the indices is arbitrary!

Task 1.3: Transpose of a Tensor

Given the Tensor $\boldsymbol{T} = T_{ij}\boldsymbol{e}_i \otimes \boldsymbol{e}_j$, determine the coefficients of its transpose $\boldsymbol{T}^T = T^T_{ij}\boldsymbol{e}_i \otimes \boldsymbol{e}_j$ using the following definition

$$\boldsymbol{w} \cdot \boldsymbol{T}\boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{T}^T\boldsymbol{w} \tag{1.1}$$

$\underline{w} = w_m \underline{e}_m$ | $\underline{\underline{T}} = T_{ij}\, \underline{e}_i \otimes \underline{e}_j$

$\underline{v} = v_n \underline{e}_n$ |

$\Rightarrow \quad \underline{w} \cdot \underline{\underline{T}}\,\underline{v} = ?$

$$\underline{\underline{T}}\,\underline{v} = (T_{ij}\underline{e}_i \otimes \underline{e}_j)(v_n \underline{e}_n)$$

$$= T_{ij} v_n \underline{e}_i (\underline{e}_j \cdot \underline{e}_n)$$

$$= T_{ij} v_n \underline{e}_i (\delta_{jn}) = T_{ij} v_j \underline{e}_i$$

$$\Rightarrow \quad \underline{w} \cdot \underline{\underline{T}}\,\underline{v} = (w_m \underline{e}_m)(T_{ij} v_j \underline{e}_i)$$

$$= w_m T_{ij} v_j (\underline{e}_m \cdot \underline{e}_i) = w_m T_{ij} v_j (\delta_{im})$$

$$= w_i T_{ij} v_j$$

$$= w_j T_{ji} v_i \quad \text{(Repeated indices can be arbitrarily renamed)}$$

→ (2)

$$\underline{v} \cdot \underline{\underline{T}}^T \underline{w}$$

$$v_n \underline{e}_n \cdot \left[\left(T^T_{ij} \underline{e}_i \otimes \underline{e}_j \right) \left(w_m \underline{e}_m \right) \right]$$

$$v_n \underline{e}_n \cdot \left[T^T_{ij} w_j \underline{e}_i \right]$$

$$= \quad v_i T^T_{ij} w_j \longrightarrow (2)$$

(1) = (2)

$$w_j T_{ji} v_i = v_i T^T_{ij} w_j$$

$$T^T_{ij} = T_{ji}$$

Task 1.4: Different Notations

Write the following expressions in symbolic and matrix notation

a) $v_i = A_{ik} w_k$,

b) $C_{ik} = A_{ij} B_{jk}$,

c) $A_{ij} B_{ji}$,

d) $u_j = A_{ik} B_{ij} v_k$,

e) $A_{ij} - \frac{1}{3}\,\delta_{ln}\,\delta_{ij}\,\delta_{km}\,\delta_{kl}\,A_{mn}$.

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a) $v_i = A_{ik} w_k$ | $\underline{v} = \underline{A}\,\underline{w}$ | $\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$

b) $C_{ik} = A_{ij} B_{jk}$ | $\underline{C} = \underline{A}\,\underline{B}$ | $\begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix}$

c) $A_{ij} B_{ji}$ | $tr(\underline{A}\,\underline{B})$ also $(\underline{A} \cdot \underline{B}^T)$ | $tr\left(\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix} \right)$

d) $u_j = A_{ik} B_{ij} v_k$ | $\underline{u} = (\underline{B}^T \underline{A}) \cdot \underline{v}$ | $\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \left(\begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{bmatrix}^T \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

e) $A_{ij} - \frac{1}{3}\,\delta_{ln}\,\delta_{ij}\,\delta_{km}\,\delta_{kl}\,A_{mn}$ $= A_{ij} - \frac{1}{3}\,\delta_{ij}\,A_{mm}$ | $\underline{A} - \frac{1}{3}\,tr(\underline{A})\,\underline{I}$ | $\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} - \frac{1}{3}(A_{11} + A_{22} + A_{33}) \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Task 1.5: Symmetric Tensors

A Tensor $\boldsymbol{S}$ is symmetric if it is equal to its transpose $\boldsymbol{S}^T$, i. e. $\boldsymbol{S} = \boldsymbol{S}^T$. For an antisymmetric/skew-symmetric Tensor $\boldsymbol{A}$ however, $\boldsymbol{A} = -\boldsymbol{A}^T$ holds true.

Each Tensor $\boldsymbol{T}$ can be additively decomposed into its symmetric and skew symmetric parts as follows

$$\boldsymbol{T} = \operatorname{sym}(\boldsymbol{T}) + \operatorname{skew}(\boldsymbol{T}), \tag{1.2}$$

with

$$\operatorname{sym}(\boldsymbol{T}) = \tfrac{1}{2}\left(\boldsymbol{T} + \boldsymbol{T}^T\right) \qquad \text{und} \qquad \operatorname{skew}(\boldsymbol{T}) = \tfrac{1}{2}\left(\boldsymbol{T} - \boldsymbol{T}^T\right). \tag{1.3}$$

Many tensors in mechanics, such as the stress tensor $\boldsymbol{\sigma}$ and the strain tensor $\boldsymbol{\varepsilon}$, are symmetric, leading to a lot of simplifications.

Given a symmetric second order tensor $\boldsymbol{S}$ and a general second order tensor $\boldsymbol{T}$, prove that

$$\boldsymbol{S} : \boldsymbol{T} = \boldsymbol{S} : \operatorname{sym}(\boldsymbol{T}). \tag{1.4}$$

⟶ Given: $\underline{S}$ is symmetric ie:

$$\underline{S}^T = \underline{S}$$

Let $\underline{S} = S_{ij}$

$\underline{T} = T_{kl}$

$\therefore \quad \underline{S} : \underline{T} = S_{ij} T_{ij}$ ⟶ (1)

$$\operatorname{Sym}(\underline{T}) = \frac{1}{2}\left(\underline{T} + \underline{T}^T\right) = \frac{1}{2}\left(T_{kl} + T_{lk}\right) \underline{e}_k \otimes \underline{e}_l$$

$$\underline{S} : \operatorname{Sym}(\underline{T}) = \left[S_{ij}\, \underline{e}_i \otimes \underline{e}_j : \frac{1}{2}\left(T_{kl} + T_{lk}\right) \underline{e}_k \otimes \underline{e}_l\right]$$

$$= \frac{1}{2}\left[S_{ij}T_{ij} + S_{ij}T_{ji}\right]$$

$$= \frac{1}{2}\left[S_{ij}T_{ij} + S_{ji}T_{ij}\right]$$ (Renamed i as j and j as i because dummy indices can be arbitrarily renamed)

$$= \frac{1}{2}\left[S_{ij}T_{ij} + S_{ij}T_{ij}\right]$$ $S_{ji} = S_{ij}$ ($\underline{S}$ is symmetric!)

$$= S_{ij}T_{ij} \rightarrow (2)$$

$\therefore$ (1) = (2) Hence proved!

Task 1.6: Volumetric–Isochoric Decomposition and some Derivatives

The strain tensor $\boldsymbol{\varepsilon}$ can be decomposed into volumetric and deviatoric parts. The volumetric part is defined by the spherical (hydrostatic) tensor

$$\operatorname{sph}[\boldsymbol{\varepsilon}] = \tfrac{1}{3}\operatorname{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I}, \tag{1.5}$$

which is the second order identity tensor multiplied by a scalar, and appears, for instance, in case of hydrostatic pressure.

The trace of the strain tensor $\boldsymbol{\varepsilon}$, which is known as the volume strain $\varepsilon_{\mathrm{V}} = \operatorname{tr}[\boldsymbol{\varepsilon}] = \varepsilon_{kk}$ is a measure of volume change during deformation.

The volume strain can be obtained from the spherical tensor ($\operatorname{tr}[\boldsymbol{\varepsilon}] = \operatorname{tr}[\operatorname{sph}\boldsymbol{\varepsilon}]$), and hence $\operatorname{sph}[\boldsymbol{\varepsilon}]$ is called the volumetric part of the strain tensor.

On the other hand, the deviatoric part of the strain tensor

$$\operatorname{dev}[\boldsymbol{\varepsilon}] = \boldsymbol{\varepsilon} - \tfrac{1}{3}\operatorname{tr}[\boldsymbol{\varepsilon}]\,\boldsymbol{I}, \tag{1.6}$$

is traceless, and it describes the volume preserving (isochoric) part of the deformation.

Evaluate the following derivatives in preparation for the next task

a) $\dfrac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\varepsilon}}$, b) $\dfrac{\partial \operatorname{tr}[\boldsymbol{\varepsilon}]}{\partial \boldsymbol{\varepsilon}}$, c) $\dfrac{\partial \operatorname{dev}[\boldsymbol{\varepsilon}]}{\partial \boldsymbol{\varepsilon}}$, d) $\dfrac{\partial \,||\operatorname{dev}\boldsymbol{\varepsilon}||}{\partial \boldsymbol{\varepsilon}}$.

a) $\dfrac{\partial \underline{\varepsilon}}{\partial \underline{\varepsilon}} = ?$ $\quad \underline{\underline{\mathbb{I}}} = \mathbb{I}_{ijkl} = \dfrac{\partial(\varepsilon_{ij})}{\partial \varepsilon_{kl}}\, \underline{e}_i \otimes \underline{e}_j \otimes \underline{e}_k \otimes \underline{e}_l$

$= \delta_{ik}\delta_{jl}\, \underline{e}_i \otimes \underline{e}_j \otimes \underline{e}_k \otimes \underline{e}_l$

b) $\dfrac{\partial \operatorname{tr}[\underline{\varepsilon}]}{\partial \underline{\varepsilon}} = \dfrac{\partial(\varepsilon_{ij}\delta_{ij})}{\partial \varepsilon_{kl}}\, \underline{e}_k \otimes \underline{e}_l = \dfrac{\partial(\varepsilon_{ii})}{\partial \varepsilon_{kl}}\, \underline{e}_k \otimes \underline{e}_l$

$= \delta_{ik}\delta_{il}\, \underline{e}_k \otimes \underline{e}_l$

$= \delta_{kl}\, \underline{e}_k \otimes \underline{e}_l = \underline{\underline{I}}$

c) $$\frac{\partial (\operatorname{dev}\underline{\varepsilon})}{\partial \underline{\varepsilon}} = \frac{\partial}{\partial \underline{\varepsilon}} \left(\underline{\varepsilon} - \frac{1}{3} \operatorname{tr}[\underline{\varepsilon}]\, \underline{I} \right)$$

$$= \delta_{ik}\delta_{jl} - \frac{1}{3}\delta_{ij}\delta_{kl}$$

$$\frac{\partial (\operatorname{dev}\underline{\varepsilon})}{\partial \underline{\varepsilon}} = \underline{\underline{\mathbb{I}}} - \frac{1}{3}\underline{I} \otimes \underline{I} = \underline{\underline{\mathbb{P}}}$$

$$\underline{\underline{\mathbb{P}}} : \underline{\varepsilon} = \operatorname{dev}\underline{\varepsilon} \quad !$$

d) $$\frac{\partial \|\operatorname{dev}\underline{\varepsilon}\|}{\partial \underline{\varepsilon}}$$

$$= \frac{\partial \|\operatorname{dev}\underline{\varepsilon}\|}{\partial (\operatorname{dev}\underline{\varepsilon})} \cdot \frac{\partial (\operatorname{dev}\underline{\varepsilon})}{\partial \underline{\varepsilon}}$$

$$= \frac{\operatorname{dev}\underline{\varepsilon}}{\|\operatorname{dev}\underline{\varepsilon}\|} : \underline{\underline{\mathbb{P}}}$$

$$= \frac{\operatorname{dev}\underline{\varepsilon}}{\|\operatorname{dev}\underline{\varepsilon}\|}$$

Note: $$\frac{\partial \|\underline{A}\|}{\partial \underline{A}} = \frac{\underline{A}}{\|\underline{A}\|}$$

$\operatorname{dev}\underline{\varepsilon} : \underline{\underline{\mathbb{P}}} = \operatorname{dev}\underline{\varepsilon}$!

Try it out as an exercise!

Revision – Voigt notation:

Voigt notation ⟶ Simply a way to represent a symmetric tensor, by reducing its order.

$$\underline{\underline{\sigma}} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \longrightarrow \text{Matrix notation}$$

$$\underline{\hat{\sigma}} = [\sigma_{11}\ \sigma_{22}\ \sigma_{33}\ \sigma_{12}\ \sigma_{23}\ \sigma_{13}\ \sigma_{21}\ \sigma_{32}\ \sigma_{31}]^T \longrightarrow \text{Reduced matrix notation}$$

$$\underline{\tilde{\sigma}} = [\sigma_{11}\ \sigma_{22}\ \sigma_{33}\ \sigma_{12}\ \sigma_{23}\ \sigma_{13}]^T \longrightarrow \text{Voigt notation}$$

Task 1.7: Matrix and Voigt Notations of Tensors

Whereas second order tensors such as the stress tensor $\boldsymbol{\sigma}$ and the strain tensor $\boldsymbol{\varepsilon}$, can be directly represented as $3\,\mathrm{x}\,3$ matrices, it is not directly achievable for higher order tensors. The relationship between $\boldsymbol{\sigma}$ und $\boldsymbol{\varepsilon}$, is given by

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}, \tag{1.7}$$

where $\mathbb{C}$ is a fourth order tensor with $3^4 = 81$ coefficients. In order to represent Equation (1.7) in matrix notation, the components of $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ can be arranged into two $9\,\mathrm{x}\,1$ and the components of $\mathbb{C}$ in a $9\,\mathrm{x}\,9$ matrix:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{21} \\ \sigma_{32} \\ \sigma_{31} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113} & \mathbb{C}_{1121} & \mathbb{C}_{1132} & \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213} & \mathbb{C}_{2221} & \mathbb{C}_{2232} & \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313} & \mathbb{C}_{3321} & \mathbb{C}_{3332} & \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213} & \mathbb{C}_{1221} & \mathbb{C}_{1232} & \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313} & \mathbb{C}_{2321} & \mathbb{C}_{2332} & \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313} & \mathbb{C}_{1321} & \mathbb{C}_{1332} & \mathbb{C}_{1331} \\ \mathbb{C}_{2111} & \mathbb{C}_{2122} & \mathbb{C}_{2133} & \mathbb{C}_{2112} & \mathbb{C}_{2123} & \mathbb{C}_{2113} & \mathbb{C}_{2121} & \mathbb{C}_{2132} & \mathbb{C}_{2131} \\ \mathbb{C}_{3211} & \mathbb{C}_{3222} & \mathbb{C}_{3233} & \mathbb{C}_{3212} & \mathbb{C}_{3223} & \mathbb{C}_{3213} & \mathbb{C}_{3221} & \mathbb{C}_{3232} & \mathbb{C}_{3231} \\ \mathbb{C}_{3111} & \mathbb{C}_{3122} & \mathbb{C}_{3133} & \mathbb{C}_{3112} & \mathbb{C}_{3123} & \mathbb{C}_{3113} & \mathbb{C}_{3121} & \mathbb{C}_{3132} & \mathbb{C}_{3131} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{32} \\ \varepsilon_{31} \end{bmatrix}. \tag{1.8}$$

Due to symmetry, the number of independent coefficients of a tensor can be reduced. For instance, a second order symmetric tensor can be uniquely defined using 6 coefficients instead of 9. This is valid for both stress $\boldsymbol{\sigma}$ and strain $\boldsymbol{\varepsilon}$ tensors. The tensor $\mathbb{C}$ also has the following symmetries

$$\mathbb{C}_{ijkl} = \mathbb{C}_{jikl} = \mathbb{C}_{ijlk}, \tag{1.9}$$

that reduce the number of its independent coefficients to 36. Additionally, its major symmetry

$$\mathbb{C}_{ijkl} = \mathbb{C}_{klij}, \tag{1.10}$$

further reduces the number of independent coefficients to 21. The aforementioned symmetries are exploited to represent the fourth order tensor $\mathbb{C}$ as a matrix using Voigt notation. Therefore, the two second order tensors $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ can be written as column vectors with 6 components, and the fourth order tensor $\mathbb{C}$ as a $6\,\mathrm{x}\,6$ matrix.

a) Write the Equation (1.8) in Voigt notation.

b) Write down the projection tensor $\mathbb{P} = \mathbb{I} - \frac{1}{3}\,\boldsymbol{I} \otimes \boldsymbol{I}$ as a $6\,\mathrm{x}\,6$ matrix using the Voigt notation, such that the deviatoric part of $\boldsymbol{\varepsilon}$ can be obtained by multiplying the aforementioned matrix with $\underline{\boldsymbol{\varepsilon}} = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{22} & \varepsilon_{33} & 2\,\varepsilon_{12} & 2\,\varepsilon_{23} & 2\,\varepsilon_{13} \end{bmatrix}^T$.

Matrix representation of $\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}$ is

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{21} \\ \sigma_{32} \\ \sigma_{31} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113} & \mathbb{C}_{1121} & \mathbb{C}_{1132} & \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213} & \mathbb{C}_{2221} & \mathbb{C}_{2232} & \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313} & \mathbb{C}_{3321} & \mathbb{C}_{3332} & \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213} & \mathbb{C}_{1221} & \mathbb{C}_{1232} & \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313} & \mathbb{C}_{2321} & \mathbb{C}_{2332} & \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313} & \mathbb{C}_{1321} & \mathbb{C}_{1332} & \mathbb{C}_{1331} \\ \mathbb{C}_{2111} & \mathbb{C}_{2122} & \mathbb{C}_{2133} & \mathbb{C}_{2112} & \mathbb{C}_{2123} & \mathbb{C}_{2113} & \mathbb{C}_{2121} & \mathbb{C}_{2132} & \mathbb{C}_{2131} \\ \mathbb{C}_{3211} & \mathbb{C}_{3222} & \mathbb{C}_{3233} & \mathbb{C}_{3212} & \mathbb{C}_{3223} & \mathbb{C}_{3213} & \mathbb{C}_{3221} & \mathbb{C}_{3232} & \mathbb{C}_{3231} \\ \mathbb{C}_{3111} & \mathbb{C}_{3122} & \mathbb{C}_{3133} & \mathbb{C}_{3112} & \mathbb{C}_{3123} & \mathbb{C}_{3113} & \mathbb{C}_{3121} & \mathbb{C}_{3132} & \mathbb{C}_{3131} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{32} \\ \varepsilon_{31} \end{bmatrix} \tag{L 2.1}$$

Due to the symmetry of $\boldsymbol{\sigma}$, this representation can be writen as

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113} & \mathbb{C}_{1121} & \mathbb{C}_{1132} & \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213} & \mathbb{C}_{2221} & \mathbb{C}_{2232} & \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313} & \mathbb{C}_{3321} & \mathbb{C}_{3332} & \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213} & \mathbb{C}_{1221} & \mathbb{C}_{1232} & \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313} & \mathbb{C}_{2321} & \mathbb{C}_{2332} & \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313} & \mathbb{C}_{1321} & \mathbb{C}_{1332} & \mathbb{C}_{1331} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \\ \varepsilon_{21} \\ \varepsilon_{32} \\ \varepsilon_{31} \end{bmatrix}.$$

Because of the symmetry of $\boldsymbol{\varepsilon}$ the last three components of $\boldsymbol{\varepsilon}$ can be omitted and the last three columns of $\mathbb{C}$ can be added to the former columns in the manner demonstrated in the following:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} + \mathbb{C}_{1121} & \mathbb{C}_{1123} + \mathbb{C}_{1132} & \mathbb{C}_{1113} + \mathbb{C}_{1131} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} + \mathbb{C}_{2221} & \mathbb{C}_{2223} + \mathbb{C}_{2232} & \mathbb{C}_{2213} + \mathbb{C}_{2231} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} + \mathbb{C}_{3321} & \mathbb{C}_{3323} + \mathbb{C}_{3332} & \mathbb{C}_{3313} + \mathbb{C}_{3331} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} + \mathbb{C}_{1221} & \mathbb{C}_{1223} + \mathbb{C}_{1232} & \mathbb{C}_{1213} + \mathbb{C}_{1231} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} + \mathbb{C}_{2321} & \mathbb{C}_{2323} + \mathbb{C}_{2332} & \mathbb{C}_{2313} + \mathbb{C}_{2331} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} + \mathbb{C}_{1321} & \mathbb{C}_{1323} + \mathbb{C}_{1332} & \mathbb{C}_{1313} + \mathbb{C}_{1331} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \end{bmatrix},$$

Due to the symmetry of $\mathbb{C}$ it can be represented as follows:

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{23} \\ \sigma_{13} \end{bmatrix} = \begin{bmatrix} \mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & 2\,\mathbb{C}_{1112} & 2\,\mathbb{C}_{1123} & 2\,\mathbb{C}_{1113} \\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & 2\,\mathbb{C}_{2212} & 2\,\mathbb{C}_{2223} & 2\,\mathbb{C}_{2213} \\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & 2\,\mathbb{C}_{3312} & 2\,\mathbb{C}_{3323} & 2\,\mathbb{C}_{3313} \\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & 2\,\mathbb{C}_{1212} & 2\,\mathbb{C}_{1223} & 2\,\mathbb{C}_{1213} \\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & 2\,\mathbb{C}_{2312} & 2\,\mathbb{C}_{2323} & 2\,\mathbb{C}_{2313} \\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & 2\,\mathbb{C}_{1312} & 2\,\mathbb{C}_{1323} & 2\,\mathbb{C}_{1313} \end{bmatrix} \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ \varepsilon_{12} \\ \varepsilon_{23} \\ \varepsilon_{13} \end{bmatrix}$$

Now the factor 2 can be shifted to the strain matrix

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{23}\\ \sigma_{13}\end{bmatrix} = \begin{bmatrix}\mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113}\\ \mathbb{C}_{2211} & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213}\\ \mathbb{C}_{3311} & \mathbb{C}_{3322} & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313}\\ \mathbb{C}_{1211} & \mathbb{C}_{1222} & \mathbb{C}_{1233} & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213}\\ \mathbb{C}_{2311} & \mathbb{C}_{2322} & \mathbb{C}_{2333} & \mathbb{C}_{2312} & \mathbb{C}_{2323} & \mathbb{C}_{2313}\\ \mathbb{C}_{1311} & \mathbb{C}_{1322} & \mathbb{C}_{1333} & \mathbb{C}_{1312} & \mathbb{C}_{1323} & \mathbb{C}_{1313}\end{bmatrix} \begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ 2\,\varepsilon_{12}\\ 2\,\varepsilon_{23}\\ 2\,\varepsilon_{13}\end{bmatrix},$$

Also due to the symmetry of $\mathbb{C}$ there are only 21 independent coefficients

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{23}\\ \sigma_{13}\end{bmatrix} = \begin{bmatrix}\mathbb{C}_{1111} & \mathbb{C}_{1122} & \mathbb{C}_{1133} & \mathbb{C}_{1112} & \mathbb{C}_{1123} & \mathbb{C}_{1113}\\ & \mathbb{C}_{2222} & \mathbb{C}_{2233} & \mathbb{C}_{2212} & \mathbb{C}_{2223} & \mathbb{C}_{2213}\\ & & \mathbb{C}_{3333} & \mathbb{C}_{3312} & \mathbb{C}_{3323} & \mathbb{C}_{3313}\\ & & & \mathbb{C}_{1212} & \mathbb{C}_{1223} & \mathbb{C}_{1213}\\ & \text{sym.} & & & \mathbb{C}_{2323} & \mathbb{C}_{2313}\\ & & & & & \mathbb{C}_{1313}\end{bmatrix} \begin{bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ 2\,\varepsilon_{12}\\ 2\,\varepsilon_{23}\\ 2\,\varepsilon_{13}\end{bmatrix}.$$

b) $\underline{\underline{\mathbb{P}}} = \underline{\underline{\mathbb{I}}} - \frac{1}{3}\,\underline{\underline{I}} \otimes \underline{\underline{I}}$

Task: write $\underline{\underline{\mathbb{P}}}$ which is a 4th order tensor as a second order tensor using the Voigt notation!

$$\underline{\underline{\mathbb{P}}} = \underbrace{\underline{\underline{\mathbb{I}}}}_{①} - \underbrace{\frac{1}{3}\,\underline{\underline{I}} \otimes \underline{\underline{I}}}_{②}$$

→ We need both ① and ② in Voigt notation!

* Compare these equations

$$\operatorname{dev}\underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}} - \frac{1}{3}\operatorname{tr}[\underline{\underline{\varepsilon}}]\,\underline{\underline{I}}$$

$$\operatorname{dev}\underline{\underline{\varepsilon}} = \underline{\underline{\mathbb{P}}} : \underline{\underline{\varepsilon}}$$

$$\underline{\underline{\mathbb{P}}} : \underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}} - \frac{1}{3}\operatorname{tr}[\underline{\underline{\varepsilon}}]\,\underline{\underline{I}}$$

$$\left(\underline{\underline{\mathbb{I}}} - \frac{1}{3}\underline{\underline{I}}\otimes\underline{\underline{I}}\right) : \underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}} - \frac{1}{3}\operatorname{tr}[\underline{\underline{\varepsilon}}]\,\underline{\underline{I}}$$

$$\underline{\underline{\mathbb{I}}} : \underline{\underline{\varepsilon}} - \frac{1}{3}\left(\underline{\underline{I}}\otimes\underline{\underline{I}}\right) : \underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}} - \frac{1}{3}\operatorname{tr}[\underline{\underline{\varepsilon}}]\,\underline{\underline{I}}$$

$$\underline{\underline{\mathbb{I}}} : \underline{\underline{\varepsilon}} = \underline{\underline{\varepsilon}} \quad !$$

and this must also hold true in the voigt notation

$$\underline{\underline{\tilde{\mathbb{I}}}} : \underline{\tilde{\varepsilon}} = \underline{\tilde{\varepsilon}}$$

We know that $\underline{\tilde{\varepsilon}} = [\varepsilon_{11}\ \varepsilon_{22}\ \varepsilon_{33}\ 2\varepsilon_{12}\ 2\varepsilon_{23}\ 2\varepsilon_{13}]^T$

and hence, we can construct $\underline{\underline{\tilde{\mathbb{I}}}}$ in voigt notation as

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 \end{bmatrix}}_{\underline{\underline{\tilde{\mathbb{P}}}}} \underbrace{\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix}}_{\underline{\tilde{\varepsilon}}} = \underbrace{\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{23} \\ 2\varepsilon_{13} \end{bmatrix}}_{\underline{\tilde{\varepsilon}}}$$

* $\underline{\underline{I}}$ in weight notation is defined as

$$\underline{\tilde{I}} = [1\ 1\ 1\ 0\ 0\ 0]^T$$

$$\therefore \underline{\tilde{I}} \otimes \underline{\tilde{I}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \underline{\tilde{P}} = \underline{\tilde{I}} - \frac{1}{3} \underline{\tilde{I}} \otimes \underline{\tilde{I}}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\underline{\tilde{P}} = \begin{bmatrix} 2/3 & -1/3 & -1/3 & 0 & 0 & 0 \\ -1/3 & 2/3 & -1/3 & 0 & 0 & 0 \\ -1/3 & -1/3 & 2/3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 \end{bmatrix}$$